



Assignment– OpenCV Practicals

Problem Statement

Implement image and video processing programs in Python using the OpenCV library to perform various operations on images and live/recorded video.

The assignment should cover:

- Reading, displaying and saving images
- Resizing images
- Flipping images (horizontal, vertical, both)
- Drawing shapes (lines, polygons) and adding text on images
- Image translation (shifting) using warpAffine
- Image rotation using getRotationMatrix2D and warpAffine
- Thresholding (Binary threshold)
- Blurring images using Gaussian Blur and Median Blur
- Morphological operations (Tophat, Blackhat)
- Edge detection using Canny
- Reading and writing video files
- Capturing live video from webcam

Use the following:

- OpenCV (cv2) library
 - NumPy for array and image manipulation
 - cv2.imshow(), cv2.waitKey() and cv2.destroyAllWindows() for displaying output
-

Expected Output

Running the Python scripts should:

- Open windows displaying the original and processed images/videos
 - Show the correct transformation/effect as per the operation performed
 - Save output files where applicable (e.g., saved images/videos)
-



Submission Guidelines:

- Create **ONE main folder**.
- Example folder name:
OpenCVPracticals-YourName
- Inside this folder include:
 - Python source code file(s) (.py / .ipynb)
 - A PDF containing complete screenshots of each practical showing code and output (if required)
- Convert the folder into **ZIP format**.
- Upload the ZIP folder on the submission portal.

If you have any doubt, you can directly contact the Tutedude mentor.